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(54) **SUBSTITUTED BENZO[F]IMIDAZO[1,2-A][1,4]DIAZEPINES AS ANESTHETICS**

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A61P 23/00 (2006.01)
C07D 487/04 (2006.01)
C07D 487/12 (2006.01)

(52) **U.S. Cl.**
CPC ***C07D 487/04*** (2013.01); ***A61P 23/00*** (2018.01)

(58) **Field of Classification Search**
CPC A61K 31/5517; C07D 487/12
USPC 514/220; 540/586
See application file for complete search history.

(56) **References Cited**

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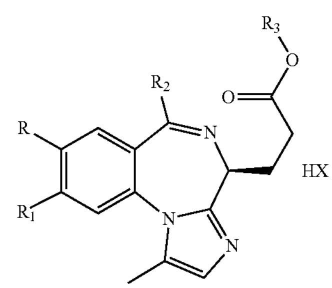
(57) **ABSTRACT**

Disclosed are a pyridyl imidazobenzodiazepine propionate (compound 1) and the synthesis and use thereof. Also provided are intermediates for the preparation. Compound 1 provided by the present invention has an obvious venous anesthesia activity equivalent to that of the positive control drug remimazolam p-toluenesulfonate or remimazolam besylate. In addition, compound 1 can be significantly reduced in mouse model experiments, and same can even overcome common limb jitters, head tilting, opisthotonus and other side effects caused by the remimazolam besylate or remimazolam p-toluenesulfonate as a drug during development in preclinical animal experiments, thereby allowing same to be used in the preparation of intravenous anesthetics. The structural general formula of compound 1 is as follows, wherein each group and substituent are as defined in the description.

1



12 Claims, 3 Drawing Sheets

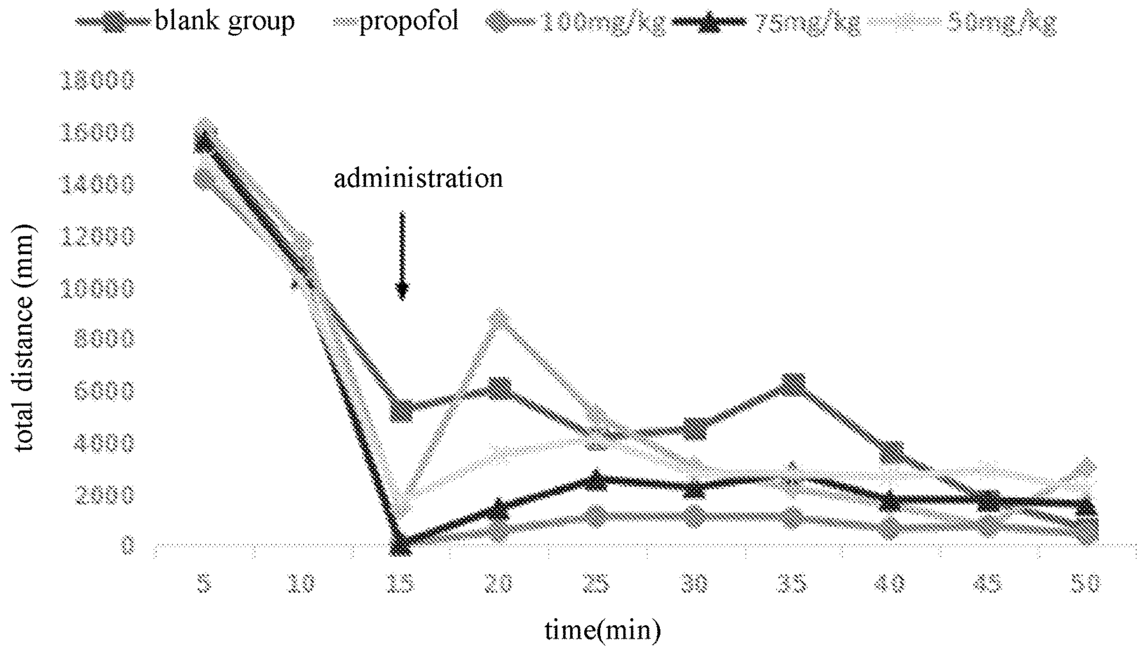


Figure 1

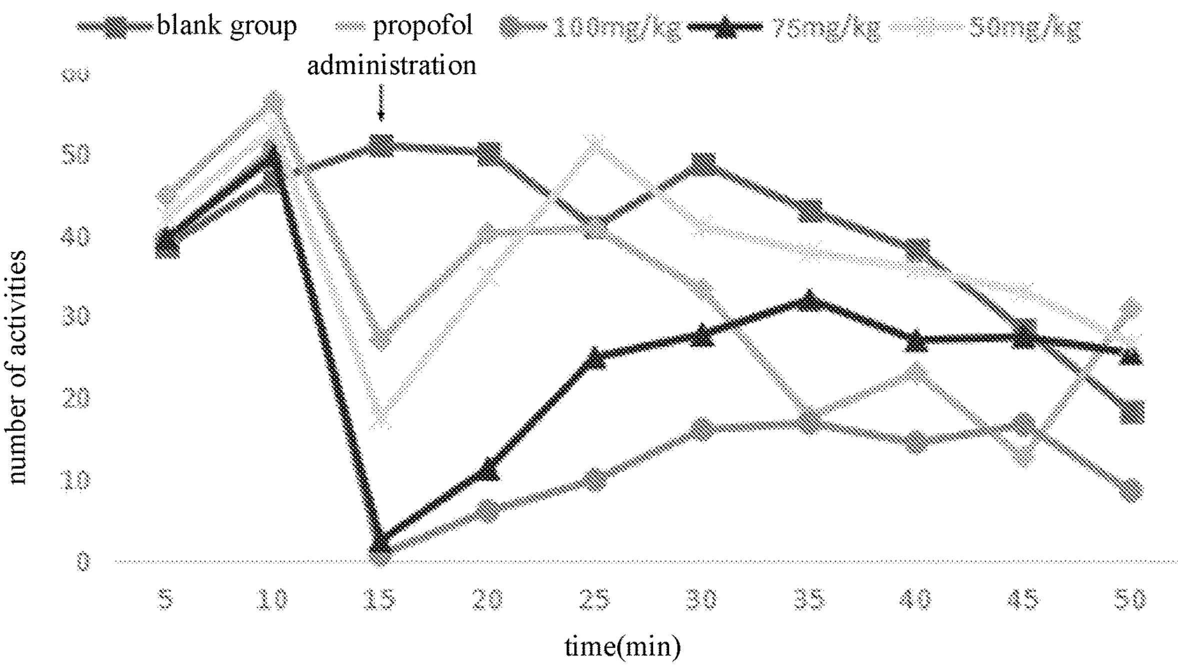


Figure 2

SUBSTITUTED BENZO[F]IMIDAZO[1,2-A][1,4]DIAZEPINES AS ANESTHETICS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a Section 371 of International Application No. PCT/CN2018/124969, filed Dec. 28, 2018, which was published in the Chinese language on Jul. 4, 2019, under International Publication No. WO 2019/129216 A1, which claims priority under 35 U.S.C. § 119(b) to CN Application No. 201711454554.9, filed Dec. 28, 2017, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention belongs to the field of medicine and chemical industry, and relates to a new pyridyl imidazobenzodiazepine propionate compound and its synthesis and application in the preparation of intravenous anesthetic drugs.

BACKGROUND ART

Remimazolam is a new type of ultra-short-acting systemic sedative anesthetic, and it is a soluble BZ derivatives, and it has a great affinity on BZ receptors in the cerebral cortex, limbic system, midbrain and brainstem spinal cord, and it can quickly and temporarily act on the 4 subtypes of γ-aminobutyric acid A (GABAA) receptor. But remimazolam has a higher affinity on the α1 subtype of the receptor, which can promote the binding of GABA and the receptor, with increased Cl^- channel opening frequency and more Cl^- influx, resulting in hyperpolarization of nerve cells, thus resulting in a nerve suppression effect. Then, it is found that Remimazolam besylate or Remimazolam p-toluenesulfonate as a drug under study commonly have side effects such as limb jitters, head tilting, opisthotonus in the preclinical animal experiments, so it is expected to develop a safer, new ultra-short-acting sedative anesthetic drugs for intravenous administration in the following clinical treatment protocols: operation of sedation, general anesthesia and ICU sedation.

SUMMARY OF THE INVENTION

The present invention provides a new pyridyl imidazobenzodiazepine propionate compound 1, which has the following structural general formula:

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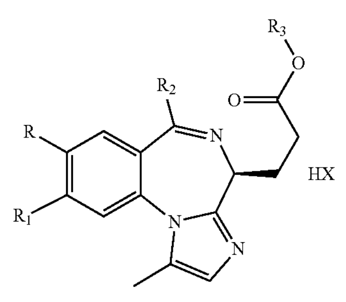

wherein, R represents various alkyl having a short carbon chain, trifluoromethyl, methoxy, nitro, fluorine, chlorine, or bromine, etc.;

R_1 represents various alkyl having a short carbon chain, trifluoromethyl, methoxy, nitro, fluorine, chlorine, or bromine, etc.;

R_2 represents a pyridine ring with nitrogen at position 2, 3 or 4;

R_3 represents various alkyl having a short carbon chain;

HX represents any acceptable pharmaceutical inorganic acids and organic acids, preferably p-toluenesulfonic acid.

In another preferred embodiment, R is selected from the group consisting of halogen, nitro, halogenated C1-C6 alkyl, C1-C6 alkyl and C1-C6 alkoxy.

In another preferred embodiment, R_1 is selected from the group consisting of halogen, nitro, halogenated C1-C6 alkyl, C1-C6 alkyl and C1-C6 alkoxy.

In another preferred embodiment, the various alkyl having a short carbon chain is C1-C6 alkyl, such as methyl, ethyl, propyl, isopropyl, butyl, isobutyl, tert-butyl, n-pentyl, isopentyl, neopentyl, n-hexyl, or a similar group.

In another preferred embodiment, the halogenated C1-C6 alkyl is a C1-C6 alkyl substituted with one or more (such as 2 or 3) halogen.

In another preferred embodiment, halogen is selected from the group consisting of F, Cl, Br and I.

In another preferred embodiment, the halogenated C1-C6 alkyl is a fluorinated C1-C6 alkyl.

In another preferred embodiment, the halogenated C1-C6 alkyl is preferably trifluoromethyl.

In another preferred embodiment, the C1-C6 alkoxy is a linear or branched alkoxy including 1-6 carbon atoms, such as methoxy, ethoxy, propoxy, isopropoxy, butoxy, isobutoxy, tert-butoxy, or a similar group.

In another preferred embodiment, R_2 is a pyridine ring with nitrogen at position 2, 3 or 4.

In another preferred embodiment, R_3 is C1-C6 alkyl.

In another preferred embodiment, HX is a pharmaceutically acceptable acid selected from the group consisting of inorganic acids, organic acids and amino acids.

In another preferred embodiment, the inorganic acids include (but are not limited to): hydrochloric acid, hydrobromic acid, hydrofluoric acid, sulfuric acid, nitric acid, or phosphoric acid.

In another preferred embodiment, the organic acids include (but are not limited to): formic acid, acetic acid, trifluoroacetic acid, propionic acid, oxalic acid, malonic acid, succinic acid, fumaric acid, maleic acid, lactic acid, malic acid, tartaric acid, citric acid, picric acid, benzoic acid, methanesulfonic acid, ethanesulfonic acid, p-toluenesulfonic acid, benzenesulfonic acid, or naphthalenesulfonic acid.

In another preferred embodiment, the organic acid is selected from the group consisting of p-toluenesulfonic acid and benzenesulfonic acid, preferably is p-toluenesulfonic acid.

In another preferred embodiment, the amino acids include (but are not limited to): citrulline, ornithine, arginine, lysine, proline, phenylalanine, aspartic acid, or glutamic acid.

In another preferred embodiment, the amino acid has a configuration selected from the group consisting of racemic configuration, D configuration, and L configuration.

In another preferred embodiment, R is selected from the group consisting of halogen, nitro, C1-C6 alkyl, halogenated C1-C6 alkyl, C3-C8 cycloalkyl, halogenated C3-C8 cycloalkyl, C1-C6 alkoxy, and halogenated C1-C6 alkoxy;

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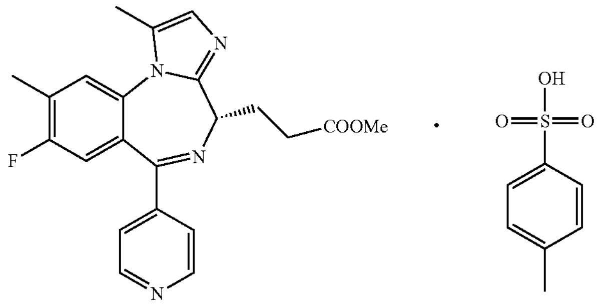
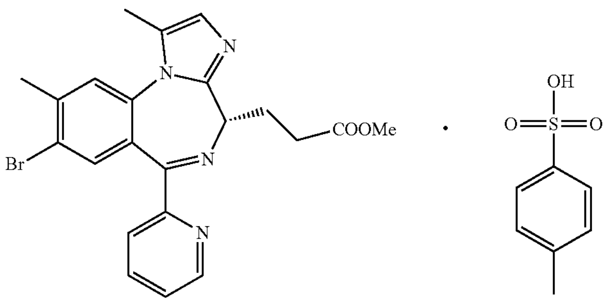
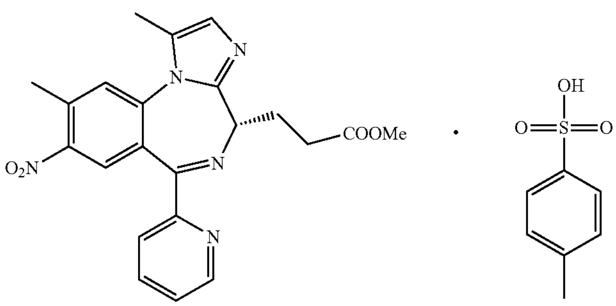
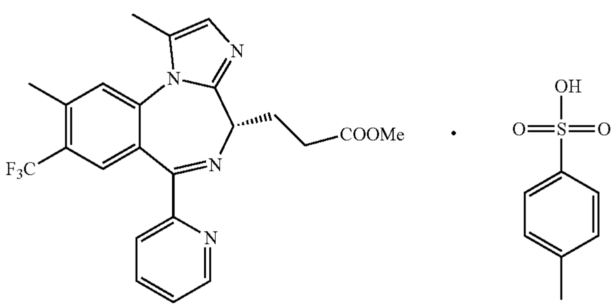
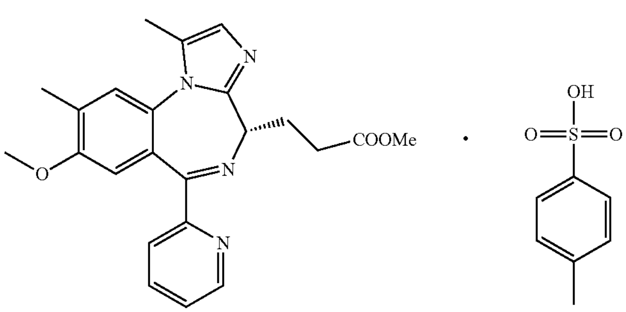
Number	Compound
1-5	
1-6	
1-7	
1-8	
1-9	

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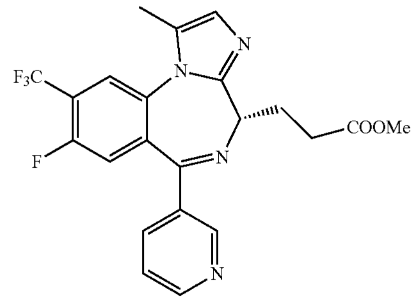
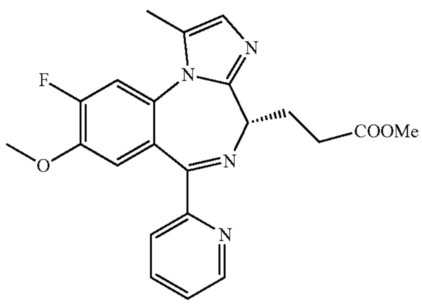
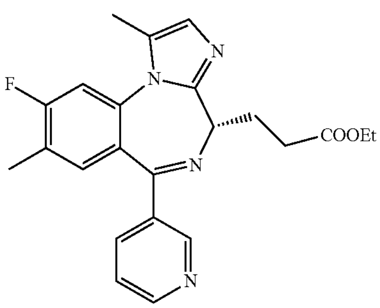
Compound




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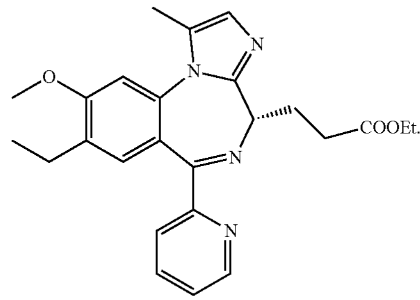
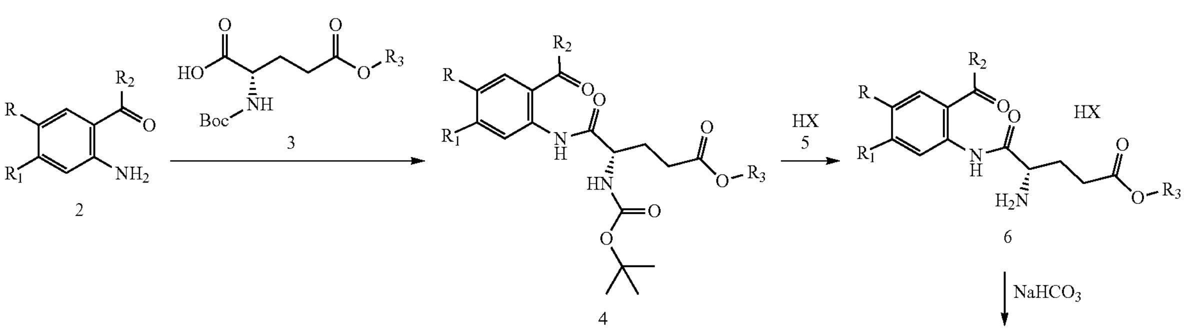
Compound


The present invention also provides a method for preparing the pyridyl imidazobenzodiazepine propionate compound 1, which is realized by the following steps:

anthranilopyridine 2 with various substituents at the 3 and 4 positions and Boc-L-glutamate-5 ester 3 are used as the starting materials to generate 2-(Boc-L-glutamate-5 esteracyl)amino-3,4-disubstituted benzoylpyridine 4 in the presence of DCC; the Boc protecting group is removed under the action of acid 5 to obtain 2-(L-glutamic acid-5esteracyl)amino-3,4-disubstituted benzoylpyridine salt 6; intramolecular condensation occurs in the presence of sodium bicarbonate to obtain 3,4-disubstituted benzodiazepine propionate 7, which then reacts wiI(R)-1-amino-2-propanol 8 to produce (S)—N-(3,4-disubstituted benzodiazepine propionate) amino-2-propanol 9; oxidative ring-closure reaction with DMP yields pyridyl imidazobenzo diazepine propionate 10; the target product pyridyl imidazobenzodiazepine propionate salt compound 1 is obtained through salt formation reaction with acid 5.

The reaction formula is as follows:



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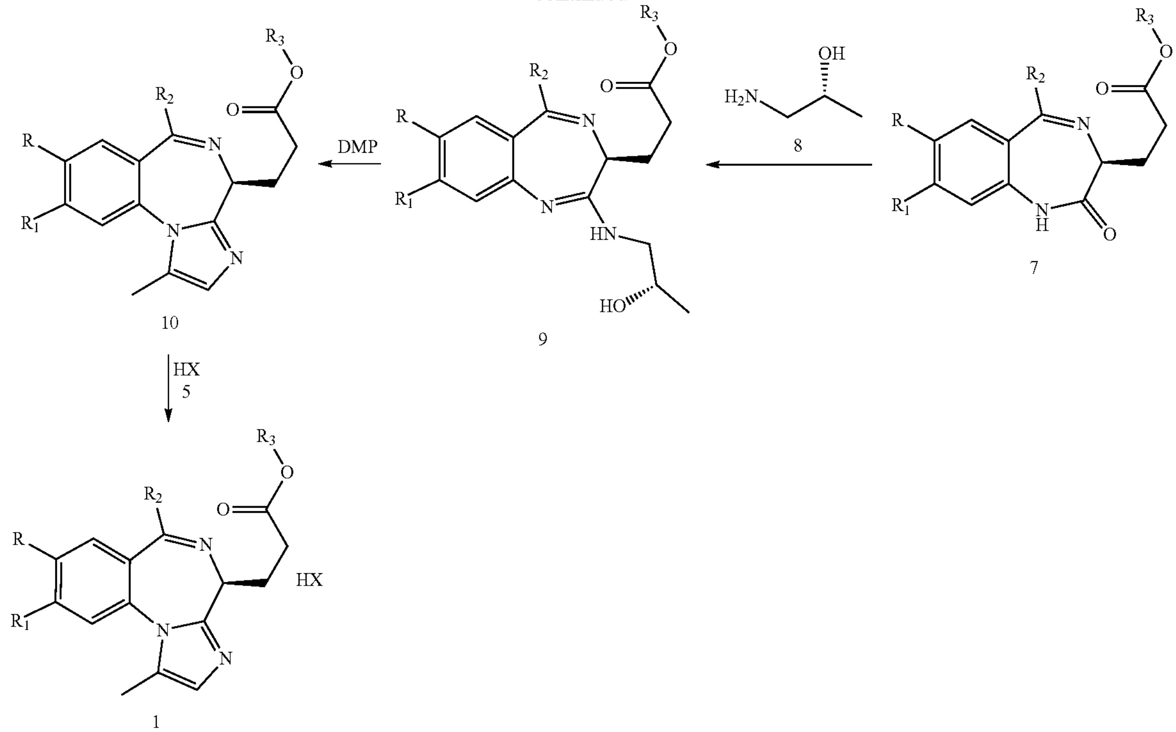


wherein, R, R_1, R_2, R_3, and HX are as described above;

DMP stands for Dess-Martin Periodinane.

The equivalent ratio of the reaction of compound 2, compound 3 and DCC is 1:0.9~1.1:1~1.5.

The equivalent ratio of the reaction of compound 6 and sodium bicarbonate is 1:10~20.

The equivalent ratio of the reaction of compounds 7 and 8 is 1:2~3.

The equivalent ratio of the reaction of compound 9 and DMP is 1:3~6, and the reaction temperature is between 30-60° C.

The equivalent ratio of the reaction of compound 10 and acid 5 in the step is 1:1-2, preferably 1:1.

HX is preferably p-toluenesulfonic acid.

The present invention also provides a pharmaceutical composition, wherein the pharmaceutical composition comprises:

1) an anesthetically effective amount of one or more of the above compound 1 and/or compound 10; and
2) a pharmaceutically acceptable carrier.

The present invention also provides a use of the pyridyl imidazobenzodiazepine propionate compound 1 for preparing intravenous anesthetic drugs.

The present invention also provides an anesthesia method, which comprises administering an anesthesia effective amount of the above compound 1 and/or compound 10 to a subject to be anesthetized.

In another preferred embodiment, the anesthesia method is used in a treatment plan selected from the group consisting of: operation of sedation, general anesthesia and ICU sedation.

The target products 1, 4, 6, 7, 9 and 10 synthesized according to the above reactions are all new compounds, and their structures were characterized by 1H NMR and ESI-MS. Through animal model test research, the results show that: compound 1 has obvious intravenous anesthesia activity, and the intravenous anesthesia activity of compound 1 is equivalent to that of the positive control drug remimazolam p-toluenesulfonate or remimazolam besylate.

In addition, the inventor surprisingly finds that, in mouse model experiments, compound 1 can significantly reduce and can even overcome common limb jitters, head ti11 pisthotonoshotonus and other side effects caused by the remimazolam besylate or remimazolam p-toluenesulfonate as a drug during development in preclinical animal experiments, so it may have the significance of further development into clinical medicine.

It should be understooI that in the Iresent invention, any of the technical features specifically described above and below (such as in the Examples) can be combined with each other, which will not redundantly be described one by one herein.

DESCRIPTION OF FIGURES

FIG. **1** shows the effect of a single administration of compound 1-2 in Example 27 on the total distance of mice's autonomous activities.

FIG. **2** shows the effect of a single administration of compound 1-2 in Example 27 on the number of mice's autonomous activities.

FIG. **3** shows the effect of a single administration of compound 1-2 in Example 27 on the rest time between mice's autonomous activities.

FIG. **4** shows the effect of multiple administrations of compound 1-2 in Example 27 on the total distance of mice's autonomous activities.

FIG. **5** shows the effect of multiple administrations of compound 1-2 in Example 27 on the rest time between mice's autonomous activities.

(2) The compound 1 and/or compound 10 significantly reduce or basically have no side-effects such as limb jitters, head tilting, 16 pisthotonos in preclinical animal experiments;

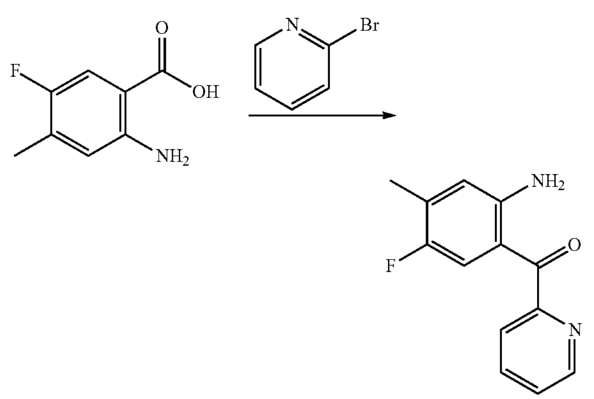
(3) The compound 1 and/or compound 10 have the characteristics of good safety, rapid onset and/or rapid recovery, that is, having very excellent comprehensive performance;

(4) The compound 1 and/or compound 10 enhance the compliance of the medication, which is reflected in that the recovery of the number of autonomous activities and the rest time of the mice after multiple administrations of compound 1-2 are better than that of propofol.

The present invention will be further illustrated below with reference to the specific examples. It should be understood that these examples are only to illustrate the invention but not to limit the scope of the invention. The experimental methods with no specific conditions described in the following examples are generally performed under the conventional conditions, or according to the manufacture's instructions. Unless indicated otherwise, parts and percentage are calculated by weight.

Unless otherwise defined, all professional and scientific terminology used in the text have the same meanings as known to the skilled in the art. In addition, any methods and materials similar or equal with the record content can be applied to the methods of the invention. The method of the preferred embodiment described herein and the material are only for demonstration purposes.

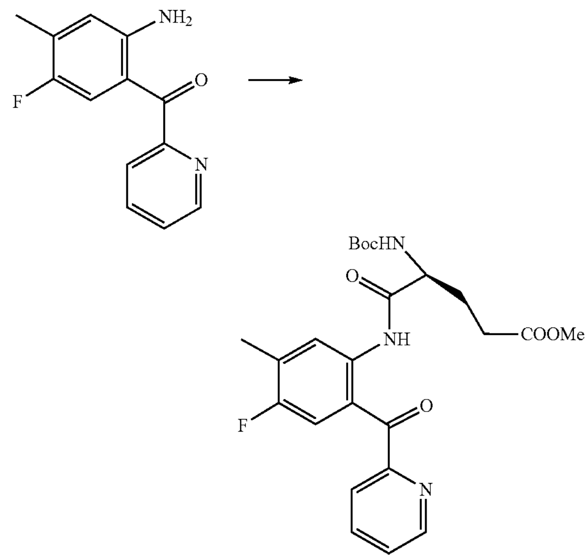
Example 1



At −40° C., 8.2 ml (85.9 mmol, 4.4 eq) of 2-bromopyridine was added to a 250 ml three-necked flask containing 31.2 ml (78.1 mmol, 4.0 eq) of n-butyllithium (2.5M) and 50 ml of anhydrous ether, and then the react solution was stirred at −40° C. for 1 hour; a solution of 3.3 g (19.53 mmol, 1.0 eq) of 2-amino-5-fluoro-4-methylbenzoic acid in 40 ml of anhydrous tetrahydrofuran was added dropwise, warmed to 0° C., and reacted for 3 hours. TLC was used to monitor the progress of the reaction. After the reaction was completed, the reaction solution was poured into 200 ml of ice water, extracted with ethyl acetate (100 ml), dried over anhydrous sodium sulfate (50 g), filtered, and the filtrate was concentrated under reduced pressure (−0.08 MPa, 40° C.) to obtain an oily matter. The residue was purified by column (ethyl acetate:petroleum ether=1:20 to 1:6) to obtain 3.59 g of yellow solid compound 1-A, with a yield of 80%.

H NMR spectrum of the compound (deuterated methanol): δ 2.33 (CH3, s, 3H), 6.96 (CH, d, H), 7.58 (CH, m, H), 7.82 (CH, m, H), 7.94 (CH, m, H), 8.10 (CH, m, H), 8.75 (CH, m, H) ppm. MS: m/z: 231.09 (M+1).

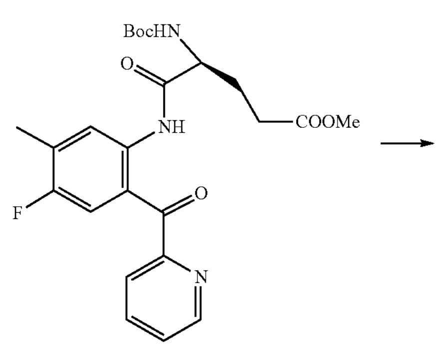
Example 2



Compound 1-A (2.64 g, 11.5 mmol, 1.0 eq) and Boc-L-glutamate-5-methyl ester (3.3 g, 12.6 mmol, 1.1 eq) were weighed into a 100 ml three-necked bottle, and 30 ml of anhydrous dichloromethane was added, dissolved with stirring, and cooled to 0° C.; 10 ml of anhydrous dichloromethane solution containing dicyclohexylimine (2.37 g, 11.5 mmol, 1.0 eq) was added dropwise. After the addition, the temperature was raised to room temperature, reacted overnight. TLC was used to monitor the progress of the reaction. After the reaction was completed, filtered, the filter cake was washed with dichloromethane (2*10 ml), and the combined filtrate was purified by column (ethyl acetate:petroleum ether=1:20 to 1:3) to obtain 5.17 g yellow thick liquid compound 1-B, with a yield of 95%.

H NMR spectrum of the compound (deuterated methanol): δ 1.42 (3CH3, s, 9H), 2.28 (CH2, m, 2H), 2.33 (CH3, m, 3H), 2.35 (CH2, m, 2H), 3.61 (CH3, s, 3H), 4.60 (CH, t, H), 7.81 (CH, d, H), 7.82 (CH, m, H), 7.83 (CH, m, H), 7.94 (CH, m, H), 8.10 (CH, m, H), 8.75 (CH, m, H) ppm. MS: m/z: 474.20 (M+1).

Example 3



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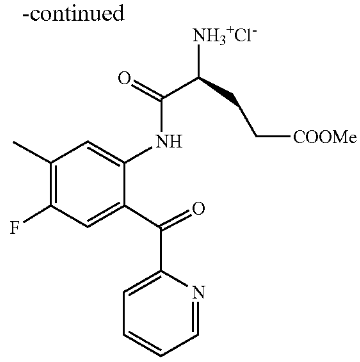
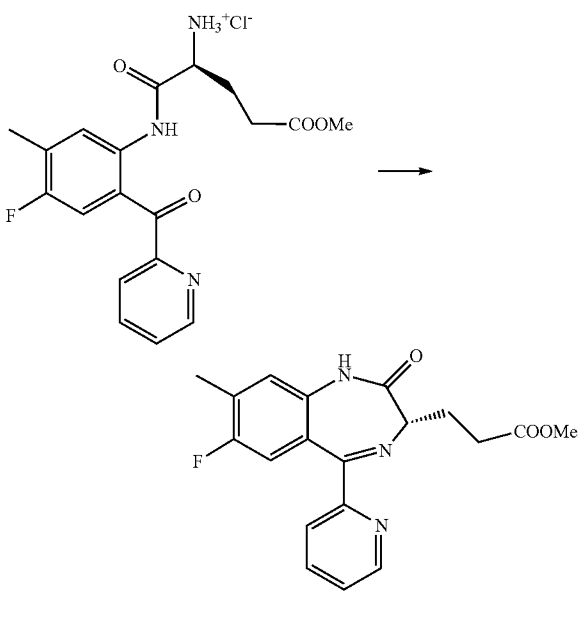


At room temperature, compound 1-B (5.6 g, 11.8 mmol, 1.0 eq) was dissolved in a 100 ml reaction flask containing 10 ml of methanol, and (17 ml, 47.4 mmol, 4.0 eq) homemade dioxane hydrochloride (2.8M) was slowly and continuously added, reacted at room temperature for 3 hours. The conversion of raw materials was complete, and the reaction solution (compound 1-C) was directly used in the next step.

H NMR spectrum of the compound (deuterated methanol): δ 2.33 (CH3, m, 3H), 2.35 (CH2, m, 2H), 2.76 (CH2, m, 2H), 3.61 (CH3, s, 3H), 4.58 (CH, m, H), 7.81 (CH, d, H), 7.82 (CH, m, H), 7.83 (CH, m, H), 7.94 (CH, m, H), 8.10 (CH, m, H), 8.31 (CH, m, H), 8.75 (CH, m, H) ppm. MS: m/z: 410.13 (M+1).

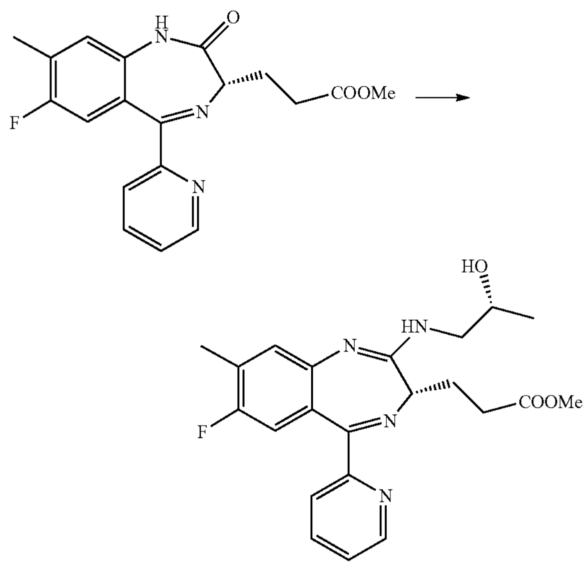
Example 4



At room temperature, sodium bicarbonate (13.9 g, 165.2 mmol, 14.0 eq) was weighed into a 100 ml reaction flask, and 37 ml of acetonitrile (methanol:dioxane:acetonitrile=3:4:10) was added, and divided into 4 portions to be added into compound 1-C reaction solution (11.8 mmol, 1.0 eq) under rapid stirring, and reacted at room temperature for 3 hours, and the conversion was complete; the reaction solution was added with diatomaceous earth (5.0 g), filtrated and concentrated under reduced pressure (−0.08 MPa, 45° C.) to obtain a pale gray solid. Residual solid was purified by column (ethyl acetate:petroleum ether=1:10 to 2:1) to obtain 3.51 g of white solid compound 1-D with a yield of 83.8% in two steps.

H NMR spectrum of the compound (deuterated methanol): δ 2.33 (CH3, m, 3H), 2.35 (CH2, m, 2H), 2.38 (CH2, m, 2H), 3.61 (CH3, s, 3H), 4.14 (CH, m, H), 7.40 (CH, d, H), 7.74 (CH, m, H), 7.79 (CH, m, H), 7.79 (CH, m, H), 7.96 (CH, m, H), 8.71 (CH, m, H), 8.83 (CH, m, H) ppm. MS: m/z: 356.13 (M+1).

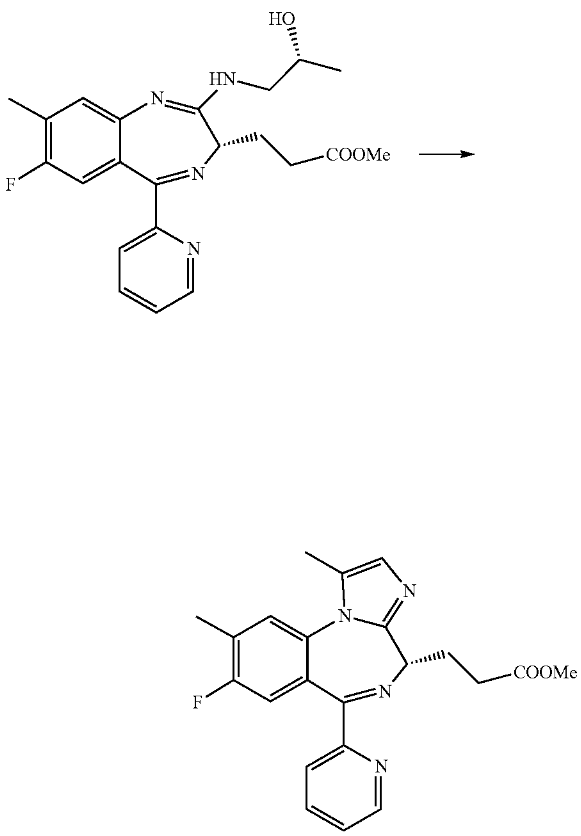
Example 5



The raw material compound 1-D (3.5 g, 9.86 mmol, 1.0 eq) was weighed into a 100 ml three-necked bottle, added with 40 ml of anhydrous tetrahydrofuran, dissolved with stirring, cooled to −18° C., under argon protection; lithium diisopropylamide (4.9 ml, 9.86 mmol, 1.0 eq) was measured and slowly and continuously added to the reaction solution, during which the temperature was controlled between −10° C. and −5° C., and then reacted at 0° C. for 0.5 h; dimorpholinophosphoryl chloride (5.0 g, 19.72 mmol, 2.0 eq) was weighed and added to the reaction solution slowly and continuously in portions over 5 minutes, and the mixture was reacted between −5° C. and 0° C. for 1 hour; ®-1-amino-2-propanol (1.8 g, 24.06 mmol, 2.44eq) was weighed to a 50 ml single-necked bottle, added with 7 ml of anhydrous tetrahydrofuran, shaked to be even, and then slowly and continuously added to the reaction solution, during which the temperature was maintained between −2° C. and 4° C., for about 40 minutes; after the addition was completed, the reaction solution was warmed to room temperature, and carried out overnight; the reaction solution was poured into ice water to be quenched, extracted with ethyl acetate, and the organic phases were combined, dried over anhydrous sodium sulfate, filtered, concentrated under reduced pressure (−0.08 MPa, 42° C.), and the residue was purified by column (ethyl acetate:petroleum ether=1:10 to pure ethyl acetate) to obtain 3.05 g of orange-yellow oily matter compound 1-E, with a yield of 75%.

H NMR spectrum of the compound (deuterated methanol): δ 1.05 (CH3, d, 3H), 2.29 (CH2, m, 2H), 2.33 (CH3, m, 3H), 2.35 (CH2, m, 2H), 3.11 (CH, m, H), 3.61 (CH3, s, 3H), 3.69; 3.44 (CH2, m, 2H), 7.05 (CH, d, H), 7.74 (CH, m, H), 7.75 (CH, m, H), 7.79 (CH, m, H), 7.96 (CH, m, H), 8.71 (CH, m, H) ppm. MS: m/z: 413.19 (M+1).

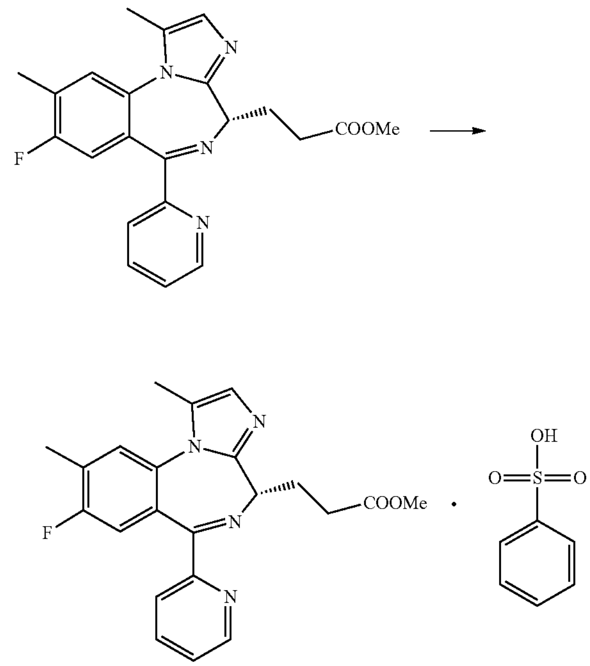
Example 6



Compound 1-E (1.5 g, 3.64 mmol, 1.0 eq) was weighed and dissolved in 14 ml butanone, warmed to 30° C. Dess-Martin oxidant (5.41 g, 12.74 mmol, 3.5eq) was added in 4 portions, then warmed to 45° C. TLC was used to monitor the progress of the reaction until the conversion of raw materials was complete; the reaction solution was quenched with saturated sodium bicarbonate solution, extracted with ethyl acetate, dried over anhydrous sodium sulfate, filtered, and the filtrate was concentrated under reduced pressure (−0.08 MPa, 45° C.). The residue was purified by the column (ethyl acetate:petroleum ether=1:10 to pure ethyl acetate) to obtain 1.12 g of brown solid compound 1-F, with a yield of 78.7%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 3.61 (CH3, s, 3H), 4.32 (CH2, m, 2H), 7.31 (CH, m, H), 7.33 (CH, m, H), 7.69 (2CH, m, 2H), 7.74 (CH, m, H), 7.75 (CH, m, H), 7.79 (CH, m, H), 7.83 (CH, m, H), 7.89 (2CH, m, 2H), 7.96 (CH, m, H), 8.71 (CH, m, H)ppm. MS: m/z: 393.16 (M+1).

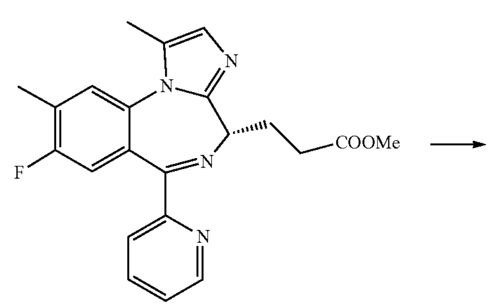
Example 7



80 mL of anhydrous ether was added to a reaction bottle containing compound 1-F (0.25 g, 0.64 mmol, 1.0 eq), dissolved with stirring, cooled to 0° C., under argon protection; 1.5 ml of anhydrous ethyl acetate containing benzenesulfonic acid (0.1 g, 0.64 mmol, 1.0 eq) was added dropwise, reacted for 1 hour, and the solid was completely precipitated. The reaction was detected by TLC to be complete and filtered. The filter cake was vacuum dried (−0.09 MPa, 20° C.) to obtain 0.31 g of off-white solid compound 1-1, with a yield of 88.9%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.31 (CH, m, H), 7.33 (CH, m, H), 7.69 (2CH, m, 2H), 7.74 (CH, m, H), 7.75 (CH, m, H), 7.79 (CH, m, H), 7.83 (CH, m, H), 7.89 (2CH, m, 2H), 7.96 (CH, m, H), 8.71 (CH, m, H)ppm. MS: m/z: 551.17 (M+1).

Example 8



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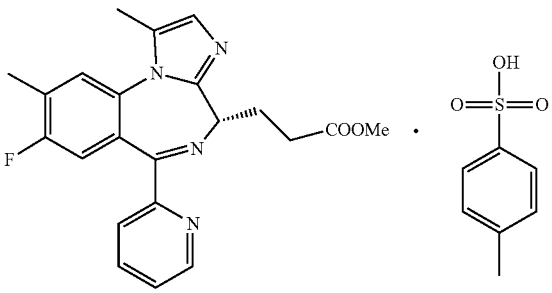
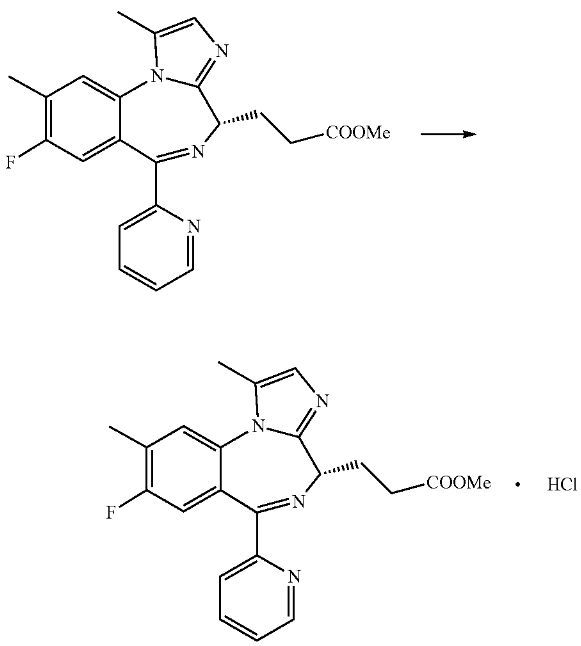


Referring to the synthesis method of Compound 1-1 in Example 7, p-toluenesulfonic acid was used as a raw material to obtain compound 1-2, with a yield of 87%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 2.43 (CH3, m, 3H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.31 (CH, m, H), 7.33 (CH, m, H), 7.47 (2CH, m, 2H), 7.74 (CH, m, H), 7.75 (3CH, m, 3H), 7.79 (CH, m, H), 7.96 (CH, m, H), 8.71 (CH, m, H) ppm. MS: m/z: 565.18 (M+1).

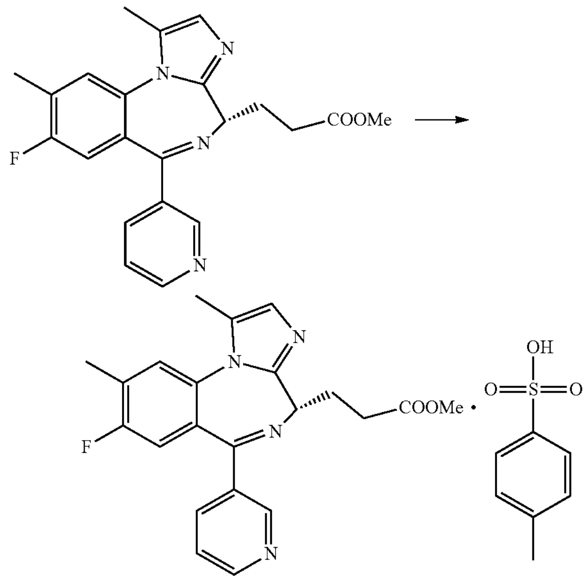
Example 9



Referring to the synthesis method of compound 1-1 in Example 7, hydrochloric acid was used as a raw material to obtain compound 1-3, with a yield of 92%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.31 (CH, m, H), 7.33 (CH, m, H), 7.74 (CH, m, H), 7.75 (CH, m, H), 7.79 (CH, m, H), 7.96 (CH, m, H), 8.71 (CH, m, H) ppm. MS: m/z: 429.14 (M+1).

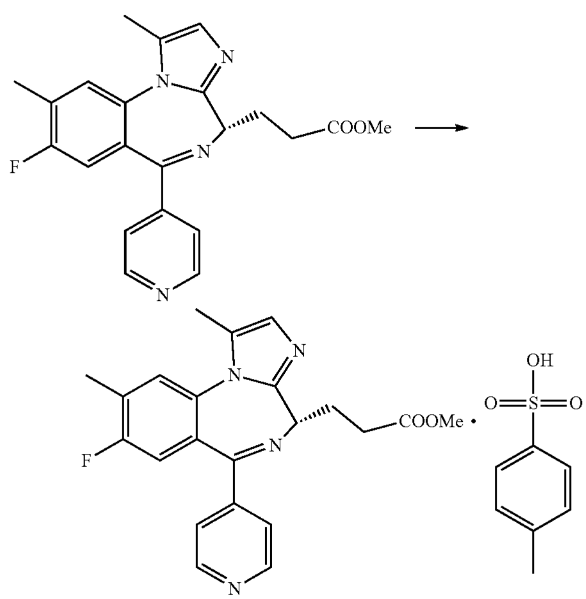
Example 10



Referring to the synthesis method of compound 1-1 in Example 7, 3-bromopyridine and p-toluenesulfonic acid were used as raw materials to obtain compound 1-4 with a yield of 85.3%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 2.43 (CH3, m, 3H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.31 (CH, m, H), 7.33 (CH, m, H), 7.47 (2CH, m, 2H), 7.58 (CH, m, H), 7.75 (3CH, m, 3H), 8.30 (CH, m, H), 8.75 (CH, m, H), 9.07 (CH, m, H) ppm. MS: m/z: 565.18 (M+1).

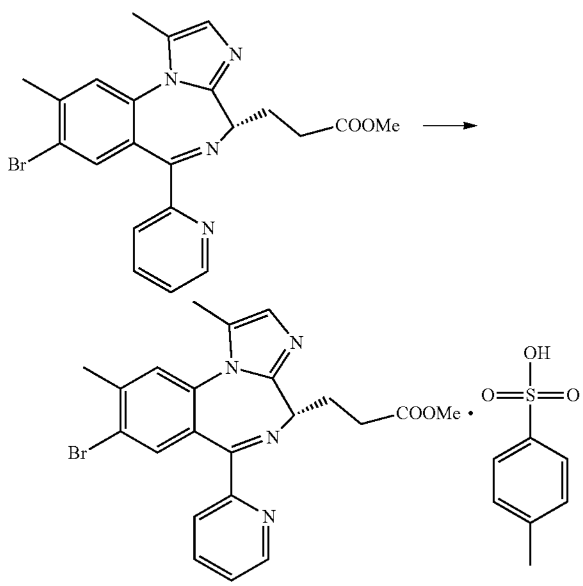
Example 11



Referring to the synthesis method of compound 1-1 in Example 7, 4-bromopyridine and p-toluenesulfonic acid were used as raw materials to obtain compound 1-5 with a yield of 86%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 2.43 (CH3, m, 3H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.31 (CH, m, H), 7.33 (CH, m, H), 7.47 (2CH, m, 2H), 7.75 (3CH, m, 3H), 7.98 (2CH, m, 2H), 8.64 (2CH, m, 2H) ppm. MS: m/z: 565.18 (M+1).

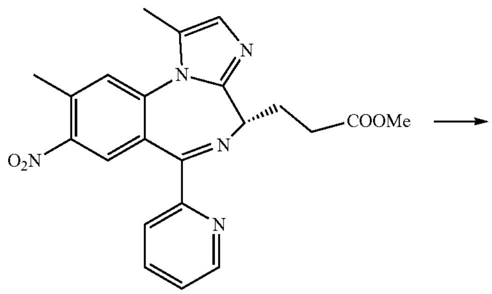
Example 12



Referring to the synthesis method of compound 1-1 in Example 7, 2-amino-5-bromo-4-methylbenzoic acid and p-toluenesulfonic acid were used as raw materials to obtain compound 1-6 with a yield of 84.5%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 2.43 (CH3, m, 3H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.24 (CH, m, H), 7.31 (CH, m, H), 7.47 (2CH, m, 2H), 7.75 (2CH, m, 2H), 7.74 (CH, m, H), 7.79 (CH, m, H), 7.86 (CH, m, H), 7.96 (CH, m, H), 8.71 (CH, m, H) ppm. MS: m/z: 625.10 (M+1).

Example 13



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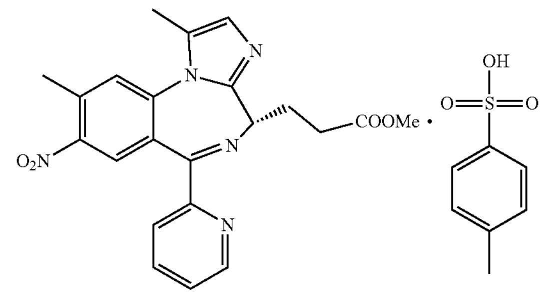
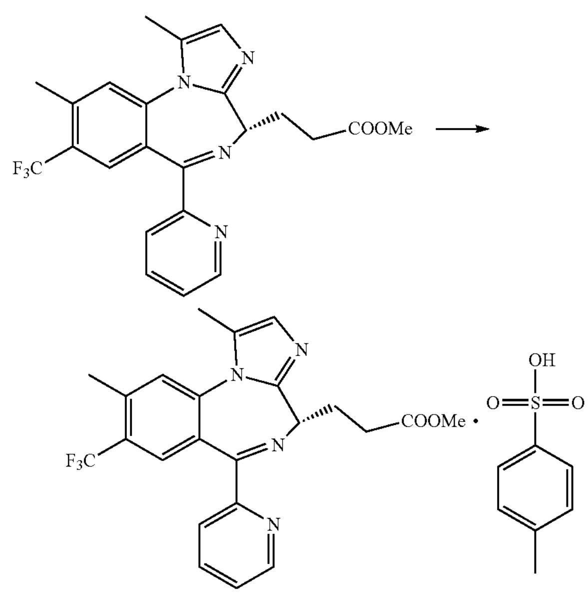


Referring to the synthesis method of compound 1-1 in Example 7, 2-amino-5-nitro-4-methylbenzoic acid and p-toluenesulfonic acid were used as raw materials to obtain compound 1-7 with a yield of 83%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.33 (CH2, m, 2H), 2.34 (CH3, m, 3H), 2.35 (CH2, m, 2H), 2.43 (CH3, m, 3H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.31 (CH, m, H), 7.47 (2CH, m, 2H), 7.75 (2CH, m, 2H), 7.61 (CH, m, H), 7.74 (CH, m, H), 7.79 (CH, m, H), 7.96 (CH, m, H), 8.47 (CH, m, H), 8.71 (CH, m, H) ppm. MS: m/z: 592.18 (M+1).

Example 14



Referring to the synthesis method of compound 1-1 in Example 7, 2-amino-5-trifluoromethyl-4-methylbenzoic acid and p-toluenesulfonic acid were used as raw materials to obtain compound 1-8 with a yield of 84.7%.

H NMR spectrum of the compound (deuterated methanol): δ 2.23 (CH3, s, 3H), 2.29 (CH3, m, 3H), 2.33 (CH2, m, 2H), 2.35 (CH2, m, 2H), 2.43 (CH3, m, 3H), 3.61 (CH3, s, 3H), 4.32 (CH, m, H), 7.28 (CH, m, H), 7.31 (CH, m, H), 7.47 (2CH, m, 2H), 7.75 (2CH, m, 2H), 7.74 (CH, m, H), 7.79 (CH, m, H), 7.96 (CH, m, H), 8.00 (CH, m, H), 8.71 (CH, m, H) ppm. MS: m/z: 615.18 (M+1).